SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
Student Name:	K Mahesh Sai	**Reg. No.:**	192412674
Section / Batch:	3rd Year	**Date:**	05-10-2026

Code of Conduct

K Mahesh Sai / 192412674 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K Mahesh Sai

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

CSA17 – ARTIFICIAL INTELLIGENCE

Assessment Tool 1 – Analytical Problem Solving

Complete Soft Copy – Questions 1 to 4

Course focus: Foundations and History of AI; Intelligent Agents; Agents and Environments; Concept of Rationality.

This write-up follows the question paper instructions: assumptions, method selection, calculations, final answers, and interpretation.

QUESTION 1: TWO-ROOM VACUUM WORLD

Topic: Agents and environments, concept of rationality

Problem understanding

A vacuum-cleaner agent operates in two squares, A and B. It starts in A, both squares are initially dirty, and dirt never reappears after cleaning. The lifetime is T = 6 time steps. At each step, the performance measure adds +1 for every clean square at the end of that step and subtracts p for every Left or Right movement. Unless otherwise stated, p = 1.

Assumptions

- At the end of each step, the number of clean squares is counted before applying the movement penalty.
- A Suck action cleans the current dirty square.
- NoOp causes no movement penalty.
- For Agent Y, once it knows both squares have been cleaned, it continues with NoOp.

(a) PEAS description – 4 Marks

PEAS component	Description
Performance Measure	+1 for every clean square at the end of each time step; −p for every Left/Right move.
Environment	Two-room vacuum world: squares A and B; both initially dirty.
Actuators	Suck, Right, Left, NoOp.
Sensors	Current location and whether the current square is Dirty or Clean.

Interpretation: The agent should clean both rooms while avoiding unnecessary movement, because movement reduces its score.

(b) Trace Agent X – 5 Marks

Method: Apply Agent X's simple-reflex rule at every time step: Dirty Suck; Clean move to the other square.

Step	Location before action	Percept	Action	Clean squares at end	Move penalty (p=1)	Step score
1	A	A Dirty	Suck	A = 1	0	1
2	A	A Clean	Right	A = 1	1	0
3	B	B Dirty	Suck	A, B = 2	0	2
4	B	B Clean	Left	A, B = 2	1	1
5	A	A Clean	Right	A, B = 2	1	1
6	B	B Clean	Left	A, B = 2	1	1

Total score:

Score(X) = 1 + 0 + 2 + 1 + 1 + 1 = 6

Therefore, for p = 1: Score(X) = 6.

Interpretation: X cleans both rooms but continues moving after both are clean, causing repeated movement penalties.

(c) Trace Agent Y – 4 Marks

Method: Apply Agent Y's reflex-with-internal-state rule. Once both squares are known to be clean, choose NoOp.

Step	Location before action	Percept / internal state	Action	Clean squares at end	Move penalty (p=1)	Step score
1	A	A Dirty	Suck	A = 1	0	1
2	A	A Clean; B not yet known clean	Right	A = 1	1	0
3	B	B Dirty	Suck	A, B = 2	0	2
4	B	Both known clean	NoOp	A, B = 2	0	2
5	B	Both known clean	NoOp	A, B = 2	0	2
6	B	Both known clean	NoOp	A, B = 2	0	2

Total score:

Score(Y) = 1 + 0 + 2 + 2 + 2 + 2 = 9

Therefore, for p = 1: Score(Y) = 9.

Y scores 9 − 6 = 3 points more than X.

(d) Total score as a function of p – 3 Marks

Agent X makes four movements. Its clean-square contribution over six steps is 10.

X(p) = 10 − 4p

Agent Y makes one movement. Its clean-square contribution is also 10.

Y(p) = 10 − p

To find when Y is strictly better:

Y(p) > X(p)

10 − p > 10 − 4p

3p > 0

Therefore, Y is strictly better when p > 0. At p = 0, both agents score 10.

(e) Which agent is rational? – 4 Marks

Agent Y is the rational agent under the given performance measure. Rationality depends on choosing an action that maximizes the expected performance based on the percept sequence and available information.

Both agents receive the same current percepts, but Y maintains an internal state. After cleaning both rooms, Y remembers that both have been cleaned and chooses NoOp. X has no internal state and therefore moves whenever its current square is clean.

Feature	Agent X	Agent Y
Architecture	Simple reflex	Reflex with internal state
Uses current percept	Yes	Yes
Uses internal state/history	No	Yes
Stops unnecessary movement	No	Yes
Score when p=1	6	9
General score	10 – 4p	10 – p

Final result: Y is better for every positive movement penalty p > 0. The performance measure and percept history make internal state valuable.

QUESTION 2: TASK ENVIRONMENT ANALYSIS AND AGENT SELECTION

Topic: PEAS, environment properties, agent types

Problem understanding

Three agents operate in different environments: E1 autonomous taxi, E2 conveyor-belt part-picking robot, and E3 tournament chess. Each of six environment properties receives 1 point for the harder value and 0 for the easier value; semi-dynamic receives 0.5.

Method

Classify each environment using the definitions in the question, assign the specified difficulty point for each property, total the six scores, then rank the environments.

(a) Classification of all six properties – 9 Marks

Environment	Observability	Agents	Determinism	Episodic / Sequential	Static / Dynamic	Discrete / Continuous
E1 – Autonomous taxi	Partially observable – traffic/intentions are not fully visible	Multi-agent – other road users act independently	Stochastic – accidents, traffic and behavior are uncertain	Sequential – current actions affect future driving states	Dynamic – road conditions change while deciding	Continuous – position, speed and steering vary continuously
E2 – Part-picking robot	Partially observable – camera has limited/noisy visual information	Single agent – robot is the acting decision-maker	Stochastic – conveyor/vision/pick outcomes can vary	Episodic – each part can be treated as a separate pick task	Dynamic – parts move while the robot operates	Continuous – position, motion and timing vary continuously
E3 – Chess	Fully observable – board state is visible	Multi-agent – two players compete	Deterministic – a legal move has a predictable board result	Sequential – each move changes the future state	Semi-dynamic – board is static but the clock changes with time	Discrete – finite board positions and legal moves

(b) Difficulty scores and ranking – 4 Marks

Property	E1 Taxi	E2 Robot	E3 Chess
Observability	1	1	0
Number of agents	1	0	1

Determinism	1	1	0
Episodic / Sequential	1	0	1
Static / Dynamic	1	1	0.5
Discrete / Continuous	1	1	0
Total difficulty score	6	4	3.5

Ranking from hardest to easiest:

1. E1 – Autonomous taxi: 6 points

2. E2 – Part-picking robot: 4 points

3. E3 – Chess: 3.5 points

Therefore, E1 is hardest and E3 is easiest under the given scoring convention.

(c) Agent architecture selection – 4 Marks

Environment	Selected architecture	Justification
Hardest: E1 Taxi	Utility-based agent	The taxi must trade off safety, travel time, traffic uncertainty and other competing outcomes. A utility function allows comparison of alternative actions under uncertainty.
Easiest: E3 Chess	Goal-based agent	Chess has a clearly defined goal: achieve a winning/checkmate state. The agent searches future states and chooses moves that help reach the goal.

Interpretation: The most difficult environment needs richer decision-making because it is partially observable, stochastic, multi-agent, sequential, dynamic and continuous.

(d) Interpretation – 3 Marks

The complexity of an agent program is determined by the properties of the environment, not simply by the application name. A taxi requires handling uncertainty, other agents, changing conditions and continuous actions. A chess system has a discrete, fully observable and deterministic board state, although it remains sequential, adversarial and time-limited. Therefore, the environment classification guides the required agent architecture and reasoning complexity.

QUESTION 3: TURING TEST DATA ACROSS THE HISTORY OF AI

Topic: Turing Test, acting humanly, eras of AI

Problem understanding

Each system faced 50 five-minute text interrogations while being the machine. The table gives the number of times the judge correctly identified the machine. Correct-identification rate = correctly identified / 50 × 100. Fooling rate = (50 – correctly identified) / 50 × 100.

Method selection

Use percentages for each system, compare correct-identification rates with the 70% reference criterion, and compare fooling rates with the 50% random-guessing level.

(a) Correct-identification and fooling rates – 6 Marks

System	Correctly identified as machine	Correct-identification rate	Taken for human	Fooling rate
ELIZA	47/50	47 ÷ 50 × 100 = 94%	3/50	3 ÷ 50 × 100 = 6%
Chatbot B	36/50	36 ÷ 50 × 100 = 72%	14/50	14 ÷ 50 × 100 = 28%
System C	29/50	29 ÷ 50 × 100 = 58%	21/50	21 ÷ 50 × 100 = 42%

Final rates: ELIZA = 94% correct, 6% fooled; Chatbot B = 72% correct, 28% fooled; System C = 58% correct, 42% fooled.

(b) Turing 70% criterion and distance from 50% chance – 5 Marks

Method: A system satisfies the stated 70% criterion if the judge's correct-identification rate is no more than 70%. Distance from chance level is measured as the absolute difference between the correct-identification rate and 50%.

System	Correct rate	Meets ≤70% criterion?	Distance from 50% chance
ELIZA	94%	No	44 percentage points above 50%
Chatbot B	72%	No	22 percentage points above 50%
System C	58%	Yes	8 percentage points above 50%

Therefore, only System C satisfies the stated 70% criterion.

(c) Improvement in fooling rate – 5 Marks

ELIZA fooling rate = 6%; System C fooling rate = 42%.

Percentage-point improvement = 42% − 6% = 36 percentage points.

Ratio = 42 ÷ 6 = 7.

Thus, System C's fooling rate is 7 times ELIZA's, an improvement of 36 percentage points.

Chatbot B to System C: 42% − 28% = 14 percentage points.

Final: ELIZA C = +36 percentage points and 7× the fooling rate; B C = +14 percentage points.

(d) Interpretation – 4 Marks

The Turing Test measures the ability to act humanly in a conversational setting. A high fooling rate indicates that judges often cannot distinguish the machine from a human under the test conditions. It does not, by itself, prove that the system is thinking, understanding or acting rationally.

Using the Chinese Room idea as an example, a system could produce convincing language responses by manipulating symbols according to rules without necessarily possessing human-like understanding. Therefore, System C's 42% fooling rate supports a claim about human-like conversational behavior, not a proof of genuine thinking or rationality.

QUESTION 4: RATIONAL ROUTE CHOICE UNDER UNCERTAINTY

Topic: Expected performance, rational versus omniscient agent

Problem understanding

The taxi must reach the airport with a 55-minute deadline. Utility is U = 100 − T − 20L, where T is travel time and L = 1 if the taxi is late, otherwise L = 0. The rational decision is the route with the highest expected utility.

Method selection

First calculate utility in each possible outcome. Then calculate expected utility using EU = Σ P(outcome) × U(outcome). Finally select the route with the highest expected utility.

(a) Utility of each route in each outcome – 4 Marks

Route	Outcome	Travel time T	Late L?	Utility U = 100 − T − 20L
R1	Normal	30 min	0 – On time	100 − 30 = 70
R1	Disrupted	70 min	1 – Late	100 − 70 − 20 = 10
R2	Normal	45 min	0 – On time	100 − 45 = 55
R2	Disrupted	60 min	1 – Late	100 − 60 − 20 = 20
R3	Normal	50 min	0 – On time	100 − 50 = 50
R3	Disrupted	50 min	0 – On time	50

(b) Expected utility and rational choice – 5 Marks

R1: Normal probability = 0.70; disruption probability = 0.30.

EU(R1) = 0.70(70) + 0.30(10) = 49 + 3 = 52.

R2: Normal probability = 0.90; disruption probability = 0.10.

EU(R2) = 0.90(55) + 0.10(20) = 49.5 + 2 = 51.5.

R3: No disruption; EU(R3) = 1.00(50) = 50.

Route	Expected utility	Rank
R1	52	1 – Highest
R2	51.5	2
R3	50	3

Therefore, the rational taxi agent should choose R1 because it has the highest expected utility: 52.

(c) New highway accident alert – 4 Marks

The percept changes the probability of R1 disruption from 0.30 to 0.80. Therefore, normal probability becomes 0.20.

EU(R1 | new alert) = 0.20(70) + 0.80(10)

= 14 + 8 = 22.

Route	Expected utility after alert	Decision
R1	22	Reject
R2	51.5	Best
R3	50	Second

New rational choice: R2, because 51.5 > 50 > 22.

(d) If the chosen route is disrupted – 4 Marks

In part (b), the rational choice was R1. If the accident actually occurs, R1 takes 70 minutes and is late.

Actual utility = 100 − 70 − 20(1) = 10.

The agent is not irrational merely because the bad outcome occurred. Rationality is judged using the information available when the decision was made, not by requiring perfect prediction.

A rational agent chooses the action expected to maximize performance based on its percept sequence. An omniscient agent would know the actual future outcome in advance. The rational agent does not have that perfect knowledge.

(e) Interpretation – 3 Marks

Rationality depends on both the percept sequence and the performance measure. The performance measure converts travel time and lateness into utility, while the percept sequence determines the probabilities and information available to the agent.

In part (c), the roads themselves did not change. However, the radio alert changed the agent's information about the highway from a 30% disruption probability to 80%. Consequently, the expected utility of R1 fell from 52 to 22, so the rational decision changed from R1 to R2.

FINAL ANSWER SUMMARY

Question	Key final results
Q1	PEAS completed; X(p) = 10 − 4p; Y(p) = 10 − p; for p=1, X=6 and Y=9; Y is strictly better for p>0; Y is rational under the stated performance measure.
Q2	Difficulty scores: E1 Taxi = 6, E2 Robot = 4, E3 Chess = 3.5. Ranking: E1 > E2 > E3. Utility-based for hardest E1; goal-based for easiest E3.
Q3	Correct-identification: ELIZA 94%, B 72%, C 58%. Fooling: 6%, 28%, 42%. Only C meets ≤70% criterion. ELIZA C: +36 percentage points and 7× ratio; B C: +14 points.
Q4	Initial expected utilities: R1=52, R2=51.5, R3=50 choose R1. After alert: R1=22 choose R2. If R1 is disrupted, actual utility=10; this does not make the earlier decision irrational.

End of Soft Copy